WORKED SOLUTION

STEP I 1997, Question 11

Mechanics • Projectile motion

STEP I 1997, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

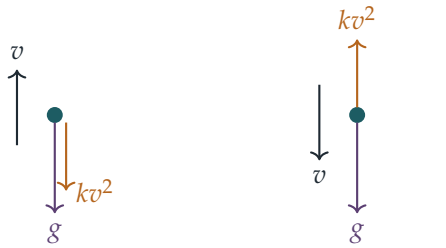
step.maths.org/questions/97-s1-q11

The key idea

The resistance depends on velocity, but we want velocity as a function of *distance*, not time. So we write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}s} = \frac{\mathrm{d}}{\mathrm{d}s}(\frac{1}{2}v^2)$. With $w = v^2$ as the unknown, the equation of motion becomes a *linear* first-order equation, which an integrating factor solves at once. The last part turns on one physical fact: resistance always opposes the motion, so it changes direction when the particle turns round.

Solution

We model the particle as a point of unit mass moving in a vertical line, acted on by its weight g (downwards) and a resistance of magnitude kv^2 directed against its velocity. Measure s upwards from the point of projection.



going up coming down

The upward motion

While the particle rises its velocity v is upwards, so the resistance kv^2 acts downwards, together with the weight. Newton's second law (mass 1) gives

$$v\frac{\mathrm{d}v}{\mathrm{d}s} = -g - kv^2,$$

using $\frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds}$. Put $w = v^2$, so that $\frac{dw}{ds} = 2v\frac{dv}{ds}$:

$$\frac{dw}{ds} + 2kw = -2g.$$

This is linear in w. Multiplying by the integrating factor e^{2ks},

$$\frac{d}{ds}(w\,e^{2ks}) = -2g\,e^{2ks} \quad \Longrightarrow \quad w\,e^{2ks} = -\frac{g}{k}\,e^{2ks} + C.$$

At $s = 0$, $w = u^2$, so $C = u^2 + \frac{g}{k}$. Hence

$$v^2 = \left(u^2 + \frac{g}{k}\right)e^{-2ks} - \frac{g}{k} = u^2e^{-2ks} + \frac{g}{k}(e^{-2ks} - 1),$$

which is $(*)$, as required. (Separating variables in $v\,dv/(g + kv^2) = -ds$ and getting a logarithm works just as well.)

Maximum height

The particle stops rising when $v = 0$; since the right-hand side of $(*)$ strictly decreases in s, this happens exactly once, at the greatest height H. Setting $v = 0$ in $(*)$:

$$\left(u^2 + \frac{g}{k}\right)e^{-2kH} = \frac{g}{k} \quad \Longrightarrow \quad e^{2kH} = 1 + \frac{ku^2}{g},$$

so

$$H = \frac{1}{2k}\ln\left(1 + \frac{ku^2}{g}\right).$$

The downward path

Ask yourself The derivation of $(*)$ never mentioned the direction of motion. Why shouldn't it carry on holding?

It did use the direction, in the sign of the resistance. On the way down the velocity is downwards, so the resistance kv^2 now acts *upwards*, against the weight. With s still measured upwards and v the upward velocity, the equation becomes

$$v\frac{dv}{ds} = -g + kv^2, \qquad \text{so} \qquad \frac{dw}{ds} - 2kw = -2g,$$

a different differential equation. So $(*)$ does not hold on the downward path.

To see this concretely, solve the new equation with $w = 0$ at $s = H$. The integrating factor e^{-2ks} gives $w = \frac{g}{k} + De^{2ks}$, and $w(H) = 0$ forces $D = -\frac{g}{k}e^{-2kH}$:

$$v^2 = \frac{g}{k}\left(1 - e^{-2k(H-s)}\right) \qquad (0 \le s \le H).$$

Using $u^2 + \frac{g}{k} = \frac{g}{k}e^{2kH}$, equation $(*)$ can be written as $v^2 = \frac{g}{k}(e^{2k(H-s)} - 1)$. Putting $x = 2k(H - s) \ge 0$, the two expressions are $\frac{g}{k}(1 - e^{-x})$ and $\frac{g}{k}(e^x - 1)$, and

$$(e^x - 1) - (1 - e^{-x}) = e^x + e^{-x} - 2 = (e^{x/2} - e^{-x/2})^2 > 0 \quad \text{for } x > 0.$$

So $(*)$ gives the wrong speed at every point below the top (they agree only at $s = H$, where both are zero); it overestimates the speed on the way down.

There is also a quick physical argument. Equation $(*)$ gives v^2 as a function of height alone, so if it held on the way down the particle would pass each height at the same speed as on the way up, and return to the start with speed u. But the resistance does negative work throughout the motion, so the particle has lost energy and must return more slowly. Indeed, at $s = 0$ the correct formula gives

$$v^2 = \frac{g}{k}\left(1 - \frac{1}{1 + ku^2/g}\right) = \frac{u^2}{1 + ku^2/g} < u^2.$$

Checks

- As $k \to 0$, $\ln(1 + ku^2/g) \approx ku^2/g$, so $H \to u^2/(2g)$, the familiar no-resistance height; and $(*)$ tends to $v^2 = u^2 - 2gs$ (expand $e^{-2ks} \approx 1 - 2ks$).
- Numerically, with $u = 10$, $k = 0.05$, $g = 9.8$, stepping the equation of motion gives $H \approx 4.1224$, matching $\frac{1}{0.1}\ln(1 + \frac{5}{9.8}) = 4.12245$; the particle returns with $v^2 \approx 66.22$, matching $u^2/(1 + ku^2/g)$, not $u^2 = 100$. Halfway down, $v^2 \approx 36.51$ while $(*)$ would give 44.87.

Common slips

- Writing the resistance as $-kv^2$ in a single equation for the whole motion: since $v^2 \geq 0$ this term always points the same way, which is exactly why one formula can't cover both legs. (A single equation would need $-kv|v|$.)
- Using $\frac{dv}{dt}$ and solving for v in terms of t, then struggling to eliminate t; $v\frac{dv}{ds}$ goes straight to the answer.

Beyond STEP

Quadratic drag is a good model for a ball moving fast through air, and the downward equation shows a *terminal velocity*: as the fall lengthens, $v^2 \to g/k$. Writing the drag as $-kv|v|$ gives one nonlinear equation for the whole motion, and a lot of first-year mechanics is about such equations, where the force depends on velocity and energy is not conserved. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces and terminal velocity carefully.

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